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ORIGINAL ARTICLE

Laser polishing of cobalt chrome alloy fabricated by laser powder bed fusion process: design of experiment-based approach for reducing surface roughness

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Abstract
The present research aims to improve the surface quality of laser powder bed fusion (LPBF) built cobalt chrome alloy (CoCr) by pulse laser remelting. The laser power, overlap percentage, and spot diameter are varied to determine the best parameters for laser polishing. The microhardness and microstructural changes associated with pulse laser remelting are described. Surface morphology and surface roughness parameters are reported for various laser remelting parameters. Among the three different spot diameters, 1, 2, and 3 mm, the maximum (> 50%) reduction in surface roughness was obtained for 2 mm with 1J laser energy and 50% *X* and 20% *Y* as well as 20% *X* and 50% *Y* overlaps. A DoE- and RSM-based approach has been utilized for parametric optimization of the surface roughness. The squared laser power is identified as a critical parameter by ANOVA. A Pareto chart confirms the significance of squared laser power across spot diameters. Optimization suggests that a 2-mm spot diameter works best with specific parameters to produce a better surface finish. However, the nanosecond order laser pulses have a limitation on the depth of surface modification without severe ablation and thereby on the reduction in the surface roughness.

Keywords Cobalt chrome · Laser powder bed fusion · Pulse laser polishing · Analysis of variance · Surface roughness

1 Introduction

The additive manufacturing (AM) of metallic materials by the laser powder bed fusion process (LPBF) has been found to be adopted by various industries due to the ease of fabrication [1]. The layer-by-layer manufacturing method could be used for making complex geometries and various high-value components. The application of LPBF has been limited to a few material systems, such as titanium alloys, cobalt-chrome alloys, Ni-base super alloys, stainless steels, aluminium alloys, refractory metals, and precious metals [2]. In as-built parts, surface roughness is typically high due to the spattering of satellite particles from melt pool plume and the morphology of solidified laser tracks, specifically on top surfaces. Laser powder bed fusion (LPBF) is prone to spatter formation due to melting pool instability and the flow field. Reza et al. compared [3] virgin and spatter powder particles, analyzing their chemical composition, phases, microstructure, morphology, and thermal behavior, and then examined the interaction of spatter particles with shielding gas and recoater motion. The surface roughness of the lateral/side surfaces is usually high compared to the top surface mainly due to the attachment of residual/unused powder particles having intimate contact with the powder bed during melt pool solidification and staircase effect (more dominant in inclined surfaces). To some degree, the amount of powder adhesion can be controlled by changing the built part orientation and powder morphology and modifying the scan pattern [4]. Mahmoodkhani et al. [5] proposed a Plackett–Burman method to identify key parameters affecting surface roughness quality in additive manufacturing of metal powders. The results show skin thickness, geometry, and layer thickness were the most significant factors affecting the surface roughness of Hastelloy X coupons printed using laser powder bed fusion. Several types of polishing methods

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can be used to enhance surface quality, with each having its distinct advantages and limitations. The conventional polishing technique is commonly used, but it has many drawbacks. Mechanical polishing is a contact-type process that can distort the surface feature, change dimensional accuracy due to material removal and residual mechanical stresses, and take a long processing time and a high processing cost; as a result, polishing may require more than one operation step. Electropolishing, though the contactless process, requires a long time to polish a certain area, and the maximum improvement in the surface finish is 50% [6]. Electrochemical methods were also used to reduce internal surface roughness in Inconel 625 laser powder-bed fusion additively manufactured parts. The study investigated the impact of electropolishing time on external surface roughness and thickness loss, resulting in an average 80% improvement [7]. On the other hand, laser polishing is a fast and contactless (mechanical force-free) thermal-based process. It is based on remelting a thin surface layer and subsequent rapid solidification. Material from the peak is redistributed to the valley through a controlled remelting and solidification sequence. Initially, when the laser beam impacts the substrate, heating and melting of mainly surface peaks and to some extent valleys occur. Thus, melted material is relocated from peaks to valleys under the multidirectional action of surface tension. Process parameters such as laser power, spot diameter, and scan rate affect the final surface roughness of the polished part [8–15].

Pulsed laser polishing is a micro-polishing technique in which a laser beam is used in pulse mode. In this process, discrete low-energy pulses irradiate the surface, thus generating a molten metal pool of depth, typically of order of 10–100 µm, and then, re-solidification occurs [10, 12]. In this process, surface tension-driven melt pool dynamics and surface flow rate which result in damped surface motion work to smooth the surface asperities. On damping out of the motion within the time when the surface is molten, a smoother surface is formed after re-solidification, thus giving better control of the melt pool depth and heat-affected zone. Jang et al. [11] differentiated laser micro-polishing from other laser polishing processes by process parameters, like the laser spot size and energy density. They used a U.V. nanosecond pulse laser to perform micro laser polishing on the metallic surface and studied the influence of laser spot scanning speed on the polishing effect, which was correlated with the laser energy density. Perry et al. [13] worked on the micro-polishing process used on surfaces obtained by micro-milling with the Ti6Al4V alloy. The laser was a pulsed (Q-switch) Nd: YAG laser with pulse durations of 600 ns and a fixed frequency of 4 kHz. Willenborg et al. [14] found that the shorter laser pulse duration transcendently influenced the shorter wavelength feature. Chao et al. [16] used a nanosecond pulsed fiber laser to polish wire–erode cutting surface of laser additive manufactured Ti alloys in which the irradiated area was 100×100 mm^2 square using hatched scanning mode with 50% overlapping and found a decrease in average roughness (Sa) from over 5 µm to less than 1 µm. The optimized laser power density was 1.20×10^7 W/cm^2, and the optimized scanning speed was 200 mm/s. Dadbakhsh et al. [17] investigated the effect of laser polishing by considering process parameters such as laser power, feed rate, and spot diameter, and found that the spot diameter was the least critical factor among the three and also showed the significance of spot diameter in the polishing process. Gora et al. [18] used a fiber laser to polish additively manufactured Ti–6Al–4V and cobalt–chrome alloys under the same polishing parameters and found better surface roughness reduction in cobalt chrome alloy. Yung et al. [19] presented a novel method for reducing surface roughness of Cobalt Chromium (CoCr) components with complex surface geometry using laser polishing. The method reduced the surface roughness by up to 93% and enhanced the mechanical hardness by 8%. The analytical model helped to understand and evaluate the underlying mechanisms of laser polishing. Steyn et al. [20] varied the laser power, defocusing, and the number of layers applied on tool steel while maintaining constant feed rate, pulse frequency, and step-over. Their experiment revealed that the amount of energy delivered by the laser beam to each point of the surface affected the quality of the final polished surface. Higher power densities and the number of layers applied essentially led to a superior surface finish. Alfieri et. al. [21] evaluated laser surface modification for improving surface topography in metal parts from selective laser melting of stainless steel. Results showed benefits of beam wobbling over linear scanning processing, as it reduced heat effects in the base metal, allowing for better optimization of processing parameters and exposure strategies. Lim et al. [22] used a Q-switch pulsed Nd: YAG laser system with a Gaussian spatial beam profile and a power of 5.5 W at 5 kHz pulse frequency to polish metastable austenitic 304 stainless steel (S.S.) by scanning the surface with a pulse overlap of 50–80% in the scanning direction and an overlap between two successive tracks of 50% and applying a power density varying from 2×10^6 to 5×10^6 W/cm^2. The experiment resulted in a 61.5% reduction in surface roughness from the initial surface roughness of 195 nm and improved corrosion resistance and microhardness. Caggiano et.al. [23] proposed an innovative laser polishing process for improving the surface finish of additive manufactured parts made of Cr–Cu precipitation hardening steel, used in the automotive industry. Brenner et al. [24] showed the feasibility of ultra-short pulsed (USP) laser polishing at 49.8MHz

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frequency and a 10 ps pulse duration and found the reduction of surface roughness from Sa = 0.41 to Sa = 0.21 µm. This process was found to be sensitive to laser fluence, and the scanning speed had only a secondary impact. Mumtaz et al. [25] reported that the laser pulse energies below 0.5 J did not produce enough heat to fully melt the powder. Pulse energies between 0.75 and 1 J produced well-formed samples that fully melted to the substrate. As pulse energies increased, above 1 J samples showed signs of excessive heat input. Thus, several investigations have shown that the optimization of process parameters such as laser energy density, spot diameter, scan speed, frequency, and pulse duration affects the final surface roughness [9–15]. The laser polishing process depends on mechanical and thermo-physical properties of the workpiece (initial roughness, homogeneity, thermal and optical properties), laser-related parameters (laser power, beam diameter, pulse duration, energy density, and focal offset distance), and scanning-related parameters (scanning speed, number, and percentage of the laser spot overlap) [2].

In the present research work, pulse laser polishing of the side (vertical) surfaces was performed with varying process parameters as: laser energy varied from 0.5 to 1 J, overlap in lateral (along scanning direction) and vertical (build-up direction) directions varied from 20 to 80% and laser beam spot diameter varied from 1 to 3 mm. Using the design of experiments (DOE) and response surface methods (RSM), the best parameters were obtained to minimize the surface roughness of LPBF CoCr alloy. The DOE-based approach to the laser polishing experiments could be useful for obtaining the major influencing parameter on surface roughness mitigation. The surface topography analysis for the best surface obtained for each laser spot diameter, microhardness study on the laser polished surface, and microstructure investigation for the maximum and minimum laser energy densities are also presented.

2 Experimental details

2.1 L-PBF specimen fabrication

Cobalt chrome alloy metal powder (Make: EOS GmbH, Germany) with powder particle size in the range of 5–50 µm was used for the fabrication of specimens [26]. EOS cobalt chrome MP1 is a cobalt-chrome-molybdenum-based super alloy powder having the chemical composition of Cr (26–30 wt%), Mo (5–7 wt%), Si (1.0 wt%), Mn (1.0 wt%), Fe (0.75 wt%), C (0.16 wt%), Ni (0.10 wt%), and balance Co. Cobalt chrome (CoCr) samples with 10 mm × 10 mm × 10 mm were produced by the L-PBF process in EOS M270 machine (Make: EOS GmbH, Germany) with the default parameters (laser power = 141 W, scan speed = 841 mm/s and hatch distance = 0.09 mm). In order to obtain almost entirely dense parts, process parameters used for the present study were selected as "EOS Direct Part" and "EOS External Support" (as supplied by the manufacturer in its software (EOSINT) settings) to build part and support structure, respectively. "EOS Direct Part" and "EOS External Support" are the licensed versions (proprietary) provided with locked preset mode with the machine. A default block-type support structure of 2 mm height designed in Materialize Magics software was used to facilitate ease of sample removal from the base plate. A nitrogen gas environment was used with 0.8 bar pressure to remove the powder contamination. SEM images of raw CoCr powder and the side surface of the as-built CoCr part are shown in Fig. 1.

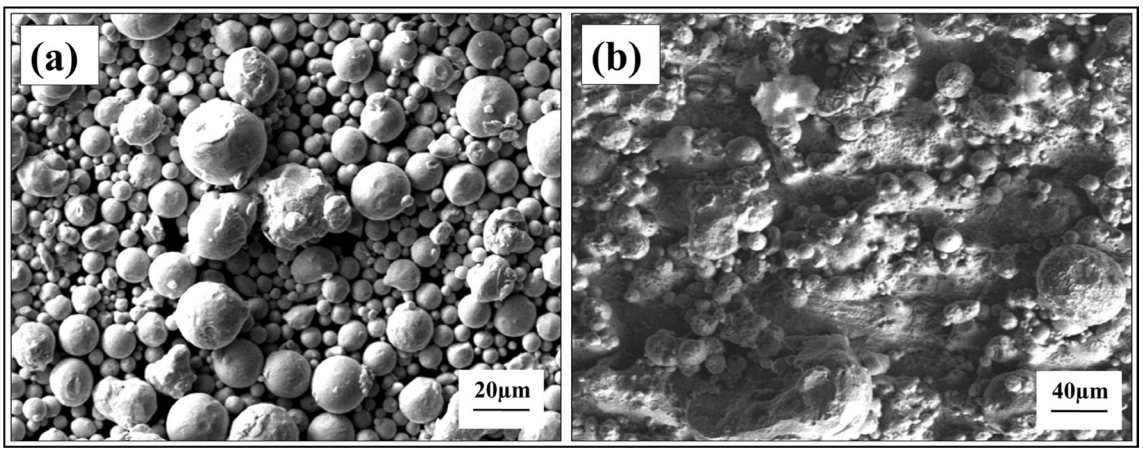


Fig. 1 SEM image of **a** raw CoCr powder and **b** as-built CoCr side surface before laser polishing

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2.2 Pulse laser polishing

Laser polishing was performed on the side surface of the LPBF cobalt chrome specimens using a Nd: YAG laser source (Make: Litron Lasers, UK) with TEM_{00} Gaussian beam profile operating at 1064 nm wavelength and having 10 ns pulse duration. Experiments were carried out at a fixed 10 Hz laser repetition frequency. The laser pulse energy was varied at three levels, 500 mJ, 750 mJ, and 1000 mJ. The laser spot diameter was varied at three levels, 1, 2, and 3 mm on the substrate. The 10 mm diameter laser beam was converged using a 500-mm focusing lens and the varying laser spot diameters were selected on the converging side of the laser beam path at 450, 400, and 350 mm distance, respectively, from the lens. The laser spot overlap (Fig. 2a) in the *X* and *Y* directions was also varied at three levels, 20%, 50%, and 80% to find out the significance of laser spot overlap in the laser polishing process. The laser polishing process was performed in ambient air conditions. The laser scan path with direction and overlap is shown in Fig. 2. The initial spiky regions of the LPBF CoCr parts would be suppressed by the laser pulses through a remelting mechanism as shown in Fig. 2b. The shallow remelting would result in reducing the surface roughness of LPBF CoCr parts. Figure 3 schematically shows the multiple times remelting in different percentage overlapped laser spot irradiation while scanning the beam along one direction.

The energy density (E.D.), in the case of continuous wave (CW) laser polishing, is the ratio of laser power to scanning speed and beam diameter. In the case of pulsed laser polishing, energy density is directly proportional to laser pulse duration, which is the interaction time between the laser and the surface. Assuming the laser beam spot diameter is circular in shape and uniform energy distribution at that spot,

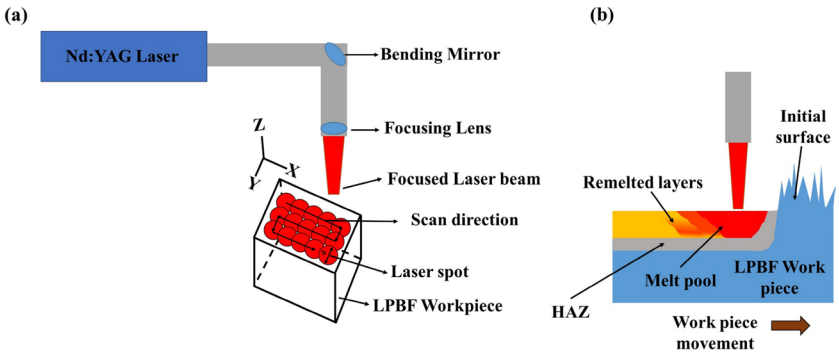


Fig. 2 Schematic diagram showing **a** Nd: YAG pulse laser polishing setup and **b** principle and mechanism of pulse laser polishing by laser-material interaction

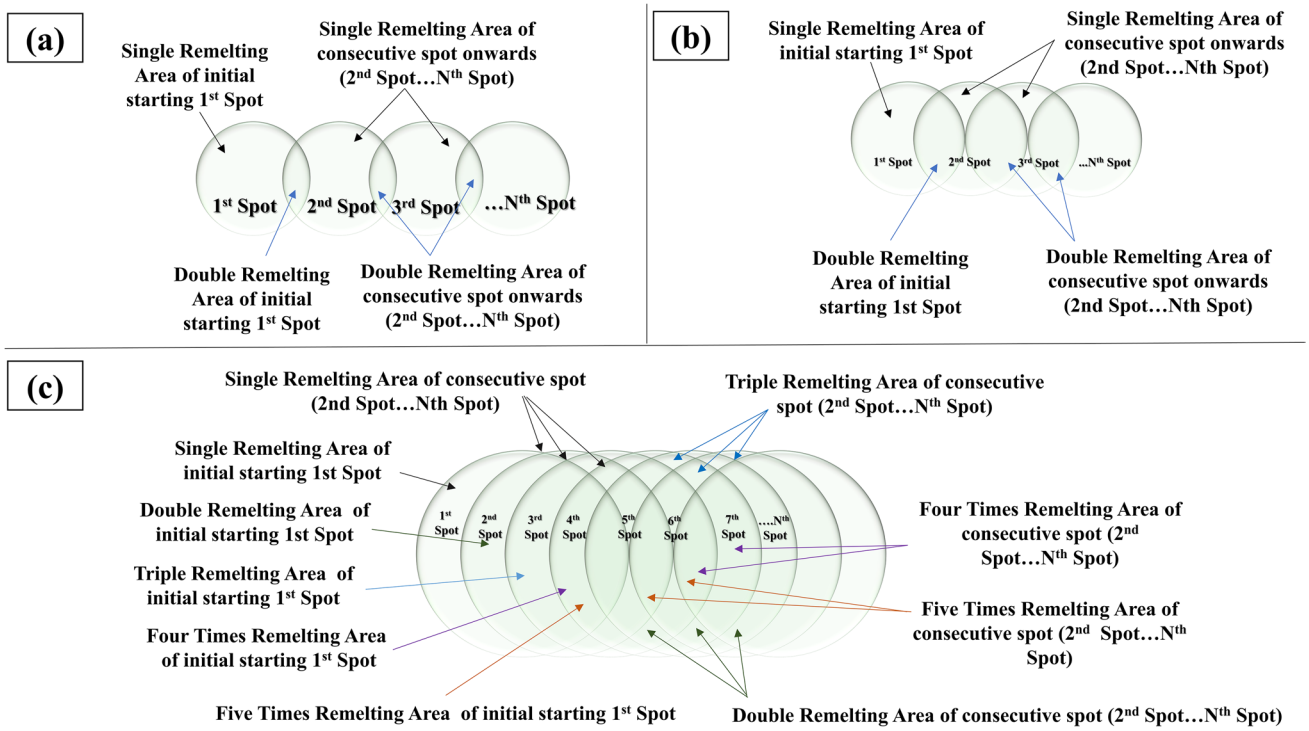


Fig. 3 Schematic of various overlaps in laser polishing. **a** 20% overlap. **b** 50% overlap. **c** 80% overlap

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energy density calculation can be done by various methods. In this paper, the energy density of a nanosecond pulsed laser has been calculated as described by Jaritngam et al. [27] and detailed calculation is provided in supplementary material S10. The process parameters used for the laser polishing of LPBF CoCr material, the corresponding number of remelt in *X* and *Y* directions and the calculated laser energy density are presented in Table 1.

2.3 Characterizations of specimens

The surface roughness of the vertical side walls of the samples was measured using a contact-type surface roughness tester (Make: Talysurf 50 (INTRA 2) 3D Profilometer, Taylor Hobson, UK) along 6 mm in *X* direction and 4 mm in Y direction. The lateral resolution which can be described along the stylus movement is 3600 points for 6mm (600

Table 1 Observation table showing initial process parameter

Sr No	Laser energy (Joule)	*X* axis overlap (%)	*Y* axis overlap (%)	N_{soX} (No. of remelt *X* axis)	N_{soY} (No. of remelt *Y* axis)	N_{SO} Net remelting ($N_{SOX}+N_{SOY}$)	Laser beam Dia. (mm)	*V* [scan speed] (mm/s)	Energy density (J/mm^2)
1	0.50	20	20	1	1	2	3	2.4	0.14
2	0.75	20	20	1	1	2	3	2.4	0.21
3	0.75	20	50	2	2	4	3	2.4	0.31
4	0.50	50	20	2	2	4	3	1.5	0.42
5	0.50	50	50	2	2	4	3	1.5	0.44
6	0.50	20	20	1	1	2	2	1.6	0.44
7	1.00	20	50	2	2	4	3	2.4	0.47
8	0.75	50	50	2	2	4	3	1.5	0.56
9	0.75	20	20	1	1	2	2	1.6	0.67
10	0.75	20	80	2	5	7	3	2.4	0.73
11	0.50	50	80	2	5	7	3	1.5	0.78
12	1.00	50	50	2	2	4	3	1.5	0.89
13	0.50	50	20	2	2	4	2	1.0	1.00
14	0.75	50	80	2	5	7	3	1.5	1.17
15	0.75	50	50	2	2	4	2	1.0	1.25
16	1.00	50	80	2	5	7	3	1.5	1.50
17	0.75	20	80	2	5	7	2	1.6	1.56
18	0.50	20	20	1	1	4	1	0.8	1.64
19	1.00	50	20	2	2	4	2	1.0	1.88
20	0.50	20	50	2	2	4	1	0.8	2.00
21	0.75	20	20	1	1	2	1	0.8	2.50
22	0.50	50	50	2	2	4	1	0.5	4.00
23	0.75	80	80	5	5	10	3	0.6	4.17
24	0.50	80	50	5	2	7	2	0.4	4.38
25	1.00	20	50	2	2	4	1	0.8	5.00
26	1.00	80	80	5	5	10	3	0.6	5.56
27	0.75	50	20	2	2	4	1	0.5	6.00
28	0.75	50	50	2	2	4	1	0.5	6.00
29	1.00	50	50	2	2	4	1	0.5	8.00
30	1.00	80	50	5	2	7	2	0.4	8.75
31	0.75	80	80	5	5	10	2	0.4	9.38
32	0.75	50	80	2	5	7	1	0.5	10.50
33	1.00	80	80	5	5	10	2	0.4	12.50
34	1.00	50	80	2	5	7	1	0.5	14.00
35	0.50	80	50	5	2	7	1	0.2	17.50
36	0.75	80	20	5	2	7	1	0.2	26.25
37	0.75	80	80	5	5	10	1	0.2	37.50
38	1.00	80	80	5	5	10	1	0.2	50.00

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points per mm in the *X* direction) and the vertical resolution is 0.05 mm/track in the *Y* direction. The resolution taken for the data acquisition is 0.05 mm/track. Contact type profilometer is considered for the current characterization because it gives fewer non-measured points and a large scan area as compared to non-contact type profilometer. The average height variation across an areal surface is quantified by the parameter areal surface roughness (Sa). It provides a measure of the overall roughness characteristics of the surface within a specific evaluation region. The root mean square height (Sq) of an areal surface is a parameter used to measure the root mean square height variance within a specific assessment region. It offers information on the surface's roughness or amplitude variations. The maximum height variation of an areal surface within a specific evaluation region is quantified by the parameter areal surface maximum height (Sz). It indicates the existence of large peaks or deep troughs in the surface texture. Sa value was obtained; after that, by first leveling the surface, Sa (leveled) was obtained. After removing the form (polynomial of degree 2) of the leveled surface, Sa (form removed) and Sa (form) were obtained [28]. In a similar way, the data for Sq and Sz has also been obtained. No filter has been used for generating the leveled, form, and form-removed surfaces in the present study. Scanning Electron Microscopy (SEM) (Make: Lyra Tescan, Czech Republic) images were taken at different magnifications to compare the microstructure of CoCr alloys. A Vickers Hardness tester (Make: Omnitech, India) was used for obtaining the microhardness. The test was performed at a load of 100 g with a dwell time of 10 s. Measurements were made at five locations of the sample, and an average was taken.

2.4 Design of experiments

The goal of conducting an experiment is to determine the impact of a given change in one or more response variables by manipulating one or more process variables or factors. The RSM method has the potential benefit of facilitating the rapid creation of a model that can reliably predict reaction times and other outputs from a set of inputs. Because of the importance of the intricate interaction between process parameters, this method is of paramount importance in such processes.

2.4.1 Response surface method

The response surface methodology (RSM) is a widely used mathematical and statistical technique for analyzing processes influenced by multiple factors with the aim of optimization. Two common variants of RSM are the box-Behnken design (BBD) and the central composite design (CCD). BBD typically analyzes at least three factors using factorial and central points, focusing on quadratic response surfaces, whereas CCD can analyze a minimum of two factors and includes axial (extreme) points along with factorial and central points, providing greater flexibility in exploring the response surface with fewer experiments. It comprises a central point and the midpoints of the circle's edges inscribed on the sphere. BBD's strength is its ability to predict complex response functions with a small number of parameters, avoiding the need for extensive experimentation. BBD was chosen over CCD for this investigation to avoid extreme values of process parameters in CCD, which would have made experimentation difficult. BBD is useful for determining the effect of process parameters on the response and its optimization. To optimize a process, it is necessary to estimate coefficients, anticipate responses, and validate the generated model [29].

Three independent input process parameters, i.e., laser energy (*A*), overlap *X*% (*B*), and overlap *Y*% (*C*) were selected to find the effect of these parameters on the surface roughness of the laser additive manufactured substrates. Based on a few pilot experimentations, different levels and ranges of the process parameters are chosen as the same for different laser beam diameters, 1 mm, 2 mm, and 3 mm. These are presented in Table 2.

3 Results and discussions

3.1 Microstructure and surface morphology of laser-polished surface

The 3D surface topography and optical and SEM images of the as-built surface and polished surfaces for processing parameters that yielded the minimum Sa values for each laser spot diameter are presented in Fig. 4 for comparison. It can be seen from the visual observation that the best surface was obtained in the case of 2 mm laser spot diameter with 1J laser pulse energy, 50% *X* overlap, and 20% *Y* overlap corresponding to 1.88 J /mm^2 laser energy density (Fig. 5). Energy density plays an important role in defining the final surface of the sample. The laser spot of 1 mm diameter with 0.75 J laser pulse energy and a relatively large overlap (50% × 50% *Y*) led to a higher energy density (6 J/mm^2), causing over-melting and burning due to oxidation as observed in Fig. 6. In the case of 3 mm spot diameter with

Table 2 Process parameters and their levels

Label	Process parameter	Unit	Low (− 1)	Middle (0)	High (+ 1)
A	Laser energy	J	0.50	0.75	1
B	Overlap *X*	%	20	50	80
C	Overlap *Y*	%	20	50	80

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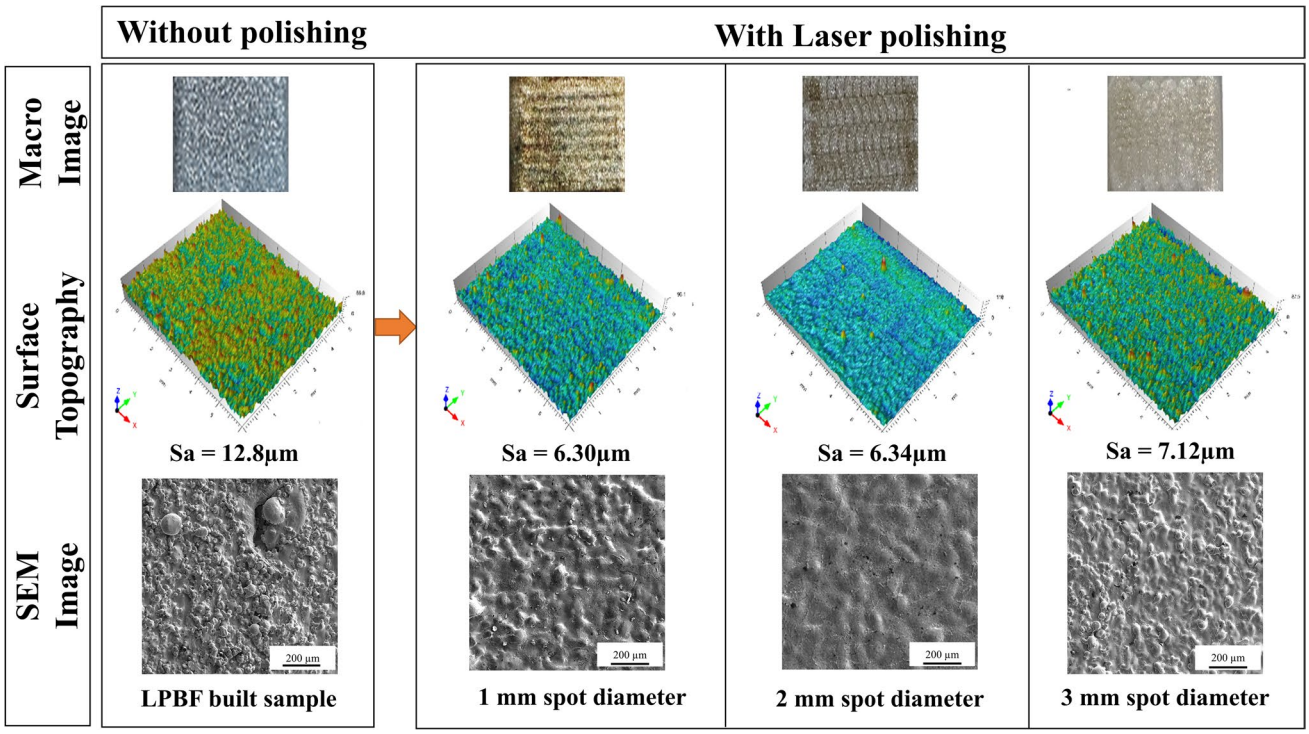


Fig. 4 Macro-image, 3D surface topography, and SEM image of the CoCr samples for unpolished and laser polished with the lowest Sa value. Laser polishing parameters (laser pulse energy, *X*% and *Y*% overlap): 1-mm spot diameter (0.75 J, 50, and 50), 2-mm spot diameter (1.0 J, 50, and 20), 3-mm spot diameter (0.75 J, 80, and 20)

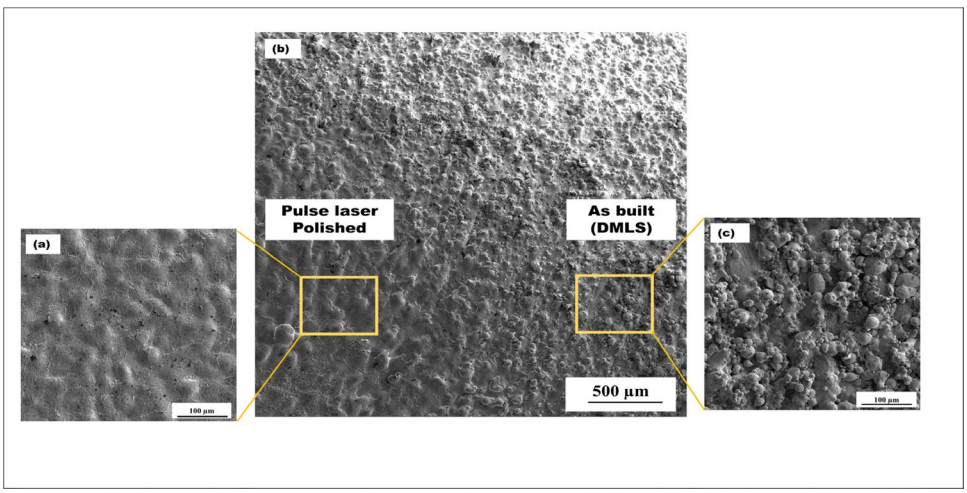


Fig. 5 SEM image showing the effect of pulse laser polishing with 2-mm spot size. **a** Polished. **b** Sample (laser energy, 1 J; 50% overlap in *X*; and 20% overlap in *Y* direction). **c** As-built

0.75 J laser pulse energy and relatively less overlap (80% *X* and 20% *Y*), since the energy density is low (0.73 J/mm^2), this could produce partial melting of the peaks (Fig. 7).

Figure 4 displays the results of pulsed laser polishing on the best surface obtained at different laser spot diameters. The initial as-built specimen had a rough surface due to un-melted powder particles that stuck from the powder bed during the laser powder bed fusion process. For a 1-mm spot diameter, as shown in the SEM image in Fig. 6, the surface appeared wavy due to improper melting of un-melted powder particles, and micro-roughness was found to be reduced slightly. For a 2-mm spot diameter, SEM image shown in Fig. 5, laser polishing produced the best surface finish with a minimum Sa of 6.34 µm (50.46% reduction from the initial Sa value of the as-built surface). For a 3-mm spot diameter, the SEM image shown in Fig. 7 showed inadequate melting of the peak, and a black oxide spot was present. The short duration (~ 10 ns) of the laser pulse limits the depth of surface melting.

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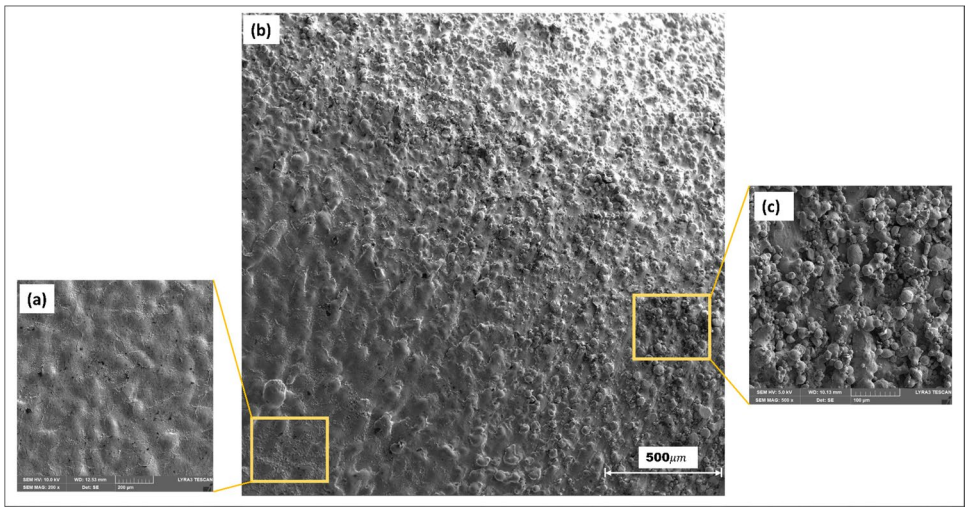


Fig. 6 SEM image showing the effect of pulse laser polishing with 1 mm spot size. **a** Polished. **b** Sample (laser energy, 0.75 J; 50% overlap in *X* and *Y* direction). **c** As-built

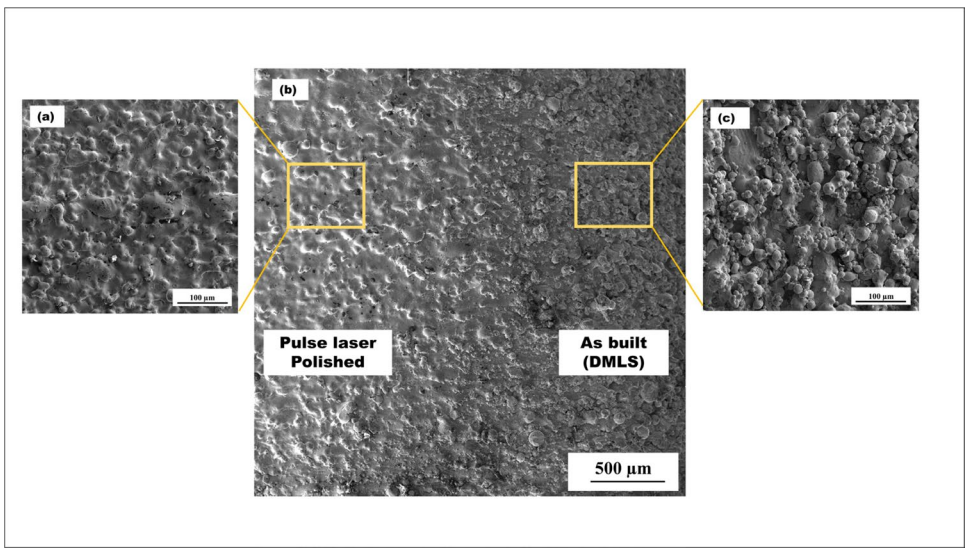


Fig. 7 SEM image showing the effect of pulse laser polishing with 3 mm spot size. **a** Polished. **b** Sample (laser energy, 0.75 J; 80% overlap in *X*; and 20% overlap in *Y* direction). **c** As-built

In the case of LPBF specimens (as-built), each layer undergoes a number of thermal cycles, influencing the solidification microstructure. A comparison of SEM images of microstructures of as-built specimen and laser-polished surfaces with the maximum (50 J/mm^2) and minimum (0.14 J/mm^2) energy densities is presented in Fig. 8. The overall microstructure shows regions consisting of columnar and cellular microstructure morphology. Microstructure of CoCrMo parts fabricated by L-PBF has been extensively studied and similar microstructure morphologies were reported [30, 31]. Even after laser micro-polishing, the microstructural features were observed. The inherent heterogeneity in the microstructure morphology and feature size could cause variations (depicted by error bars) in the microhardness, as depicted in Fig. 9.

3.2 Microhardness analysis

The surface microhardness of the base material (as-built) and the specimens subjected to varying energy density was measured and the values of the polished surfaces are plotted as a function of energy density in Fig. 9. The base material (as-built) microhardness was measured to be 480 ± 24 HV. It is observed that there is no specific trend of variation in microhardness of the polished surface with energy density, it varies within a relatively broad range, and the average values for some of the energy densities are lower and for some other energy densities are higher than that of the base material. Microhardness is affected by the heterogeneity of microstructure, formation of hard phases, grain refinement, and formation of defects, like porosity during laser polishing. The thermal

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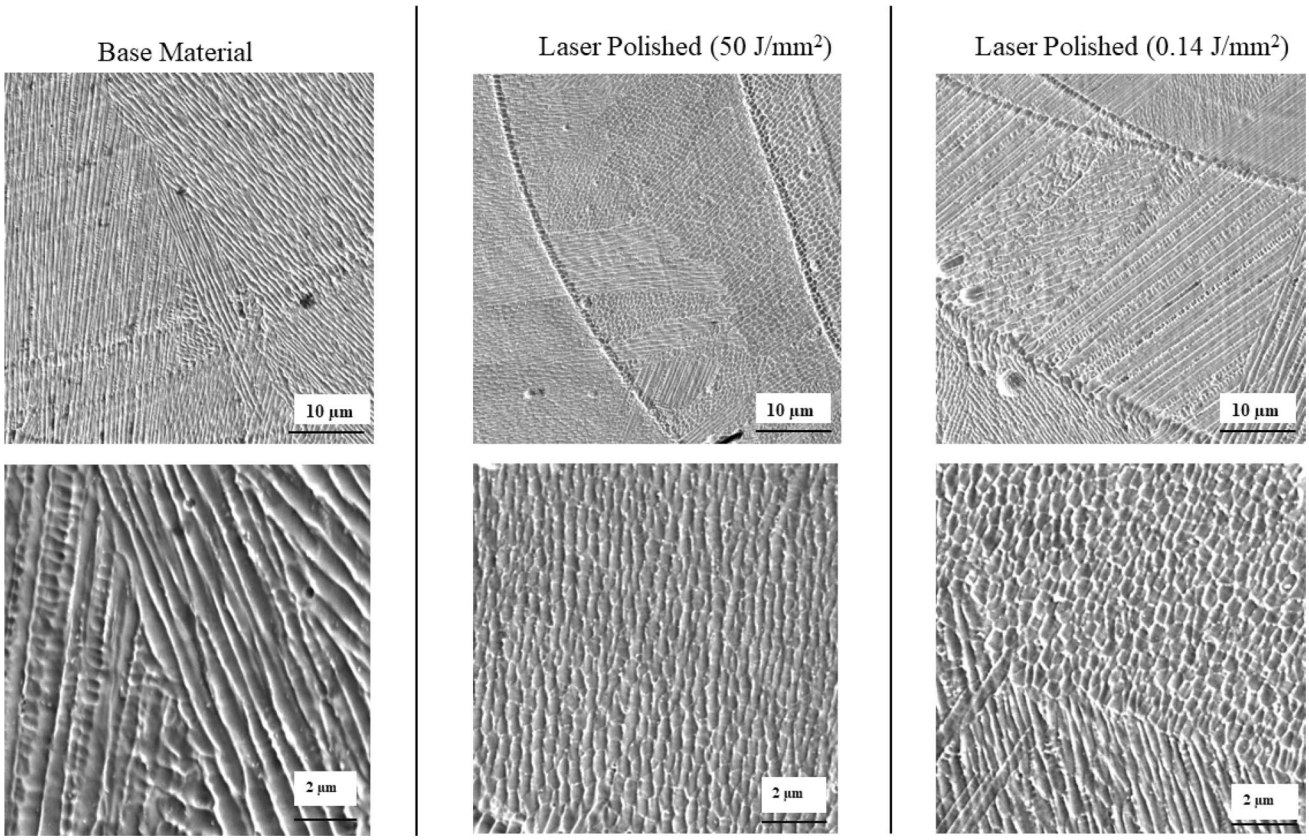


Fig. 8 Microstructure of base material and laser polished parts with maximum (50 J/mm^2) and minimum (0.14 J/mm^2) energy density

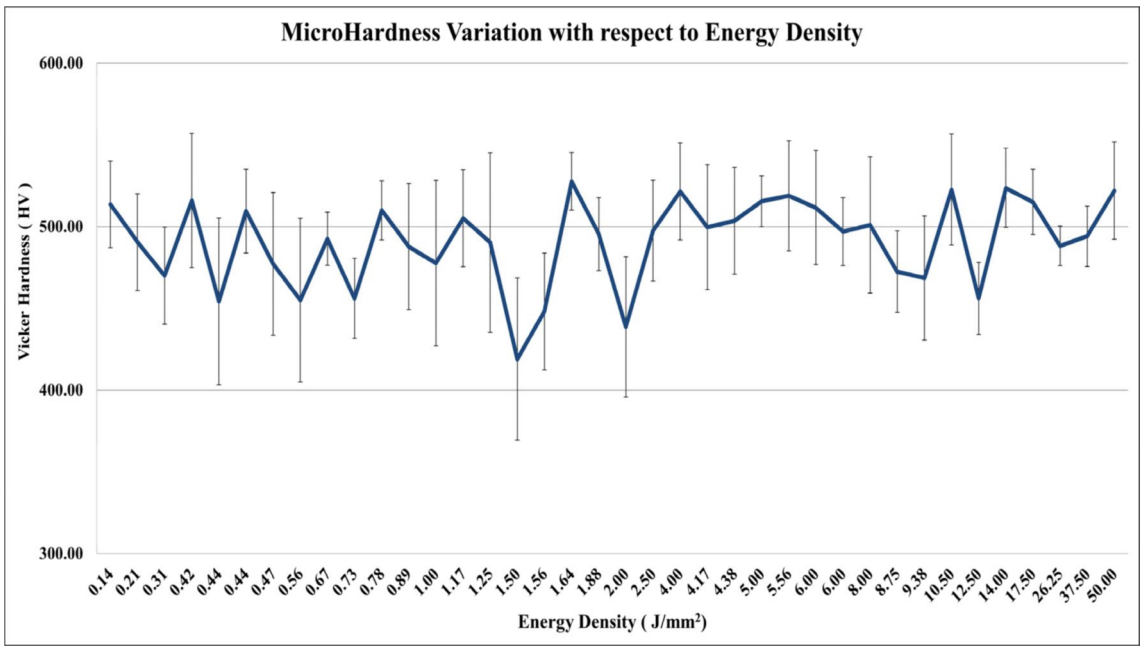


Fig. 9 Variation of microhardness with respect to the energy density and corresponding standard deviation

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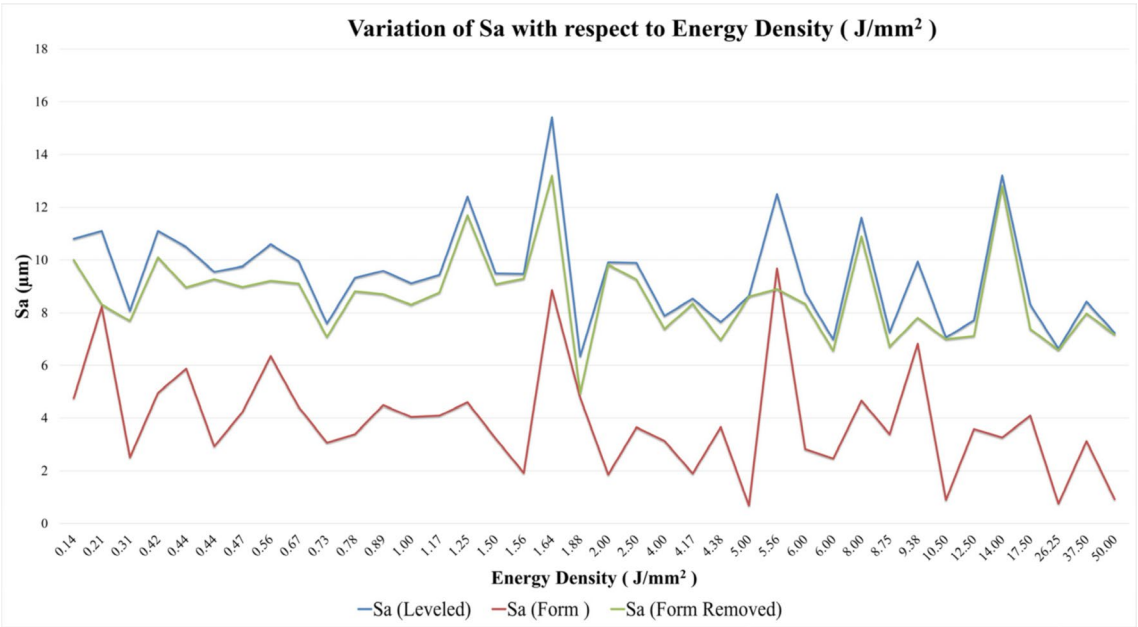


Fig. 10 Surface roughness Vs. energy density for Sa plot of the laser polished specimens

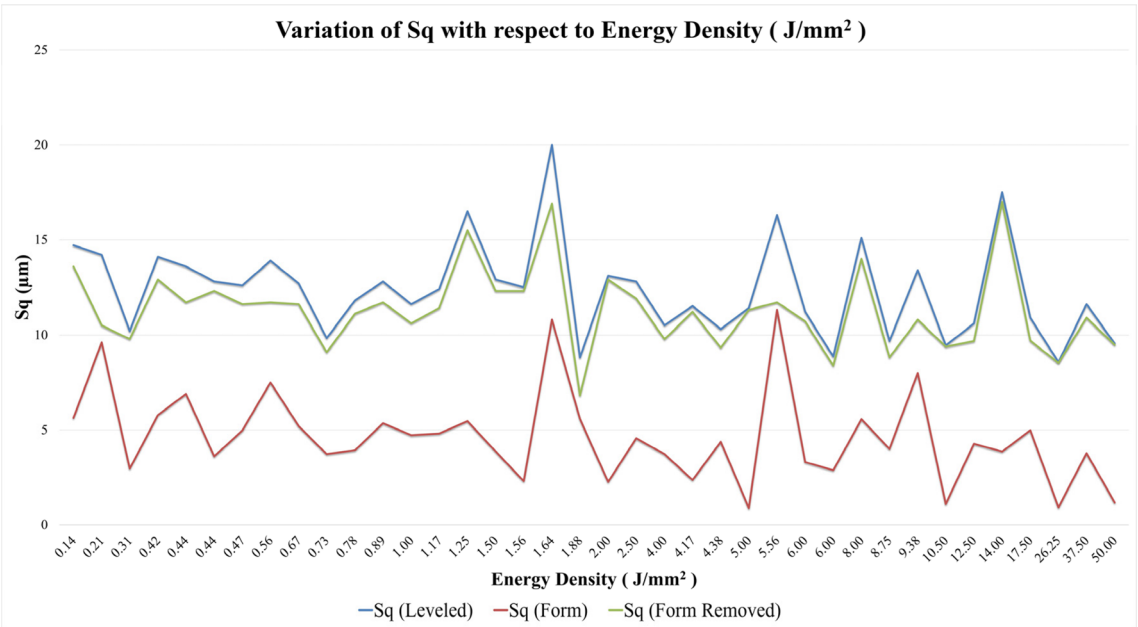


Fig. 11 Surface roughness Vs. energy density for Sq plot of the laser polished specimens

history of solidification, i.e., molten pool lifetime, temperature gradient, and cooling rate, influences these factors.

3.3 Surface roughness analysis

In the initial surface roughness, Sa of the as-built LPBF CoCr side surface was found to be 12.8 µm. The variation of Sa (leveled), Sa (form), and Sa (form removed) for different laser energy densities (Table 1) is shown in Fig. 10. Similar variations of Sq and Sz are presented in Figs. 11 and 12, respectively. The form in the sample was found in the initial sample after L-PBF. Change in form was minimal in all cases, which depicts laser polishing being performed in a low pulse energy range. The process primarily reduces

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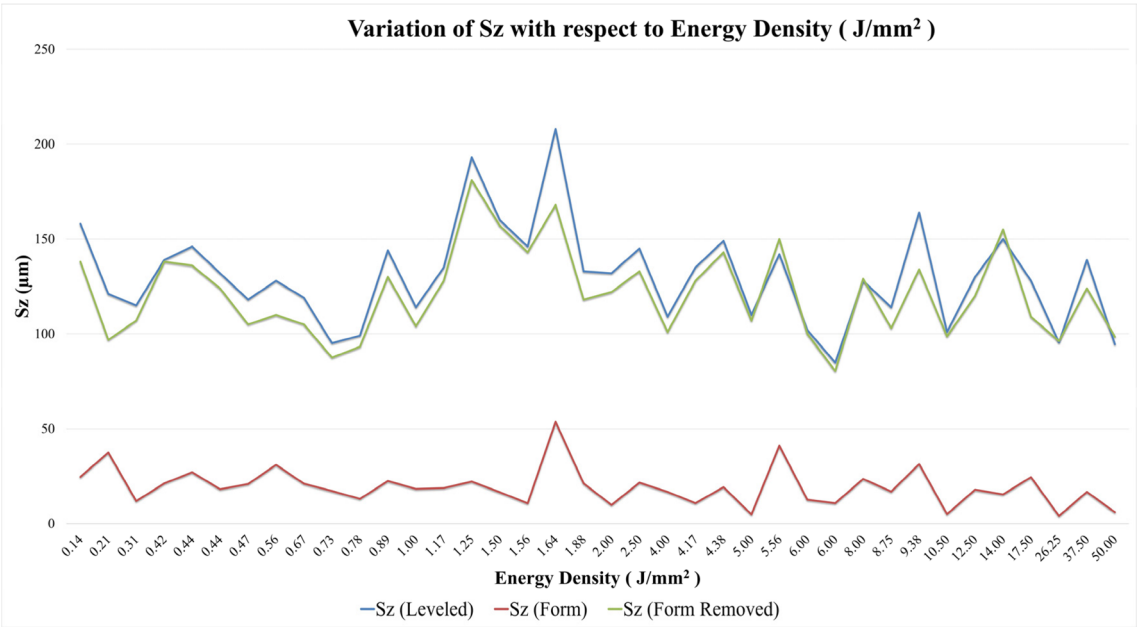


Fig. 12 Surface roughness Vs. energy density for Sz plot of the laser polished specimens

roughness value inducing minimum form and waviness, thus confirming the micro-polishing process by the pulsed laser. For each laser spot diameter, variations of Sa (leveled), Sa (form), Sa (form removed), Sq (leveled), Sq (form), Sq (form removed), Sz (leveled), Sz (form), and Sz (form removed) with energy density are shown in figures from S1 to S9 presented in supplementary data. The surface roughness values varied up and down with a little downswing with increasing energy density. The two Sa peaks exceeding the initial surface roughness in Fig. 10 at 1.68 and 14 J/mm^2 energy densities are produced by a 1-mm laser beam. The 1-mm diameter laser pulse has a power density of more than 50 MW/mm^2 (Table 1) and can cause over-melting of peak and valley both partially ablated, similar to pulse laser drilling [32]. The peak Sa at 5.56 J/mm^2 energy density, though below the initial surface roughness, is resulted by the 3-mm laser spot diameter having 14 MW/ mm^2 power density and this could be due to the partial remelting. For the 2-mm spot

Table 3 Experimental table with output responses Sa (leveled)

Run	Laser energy	Overlap *X*%	Overlap *Y*%	Sa (1 mm)	Sa (2 mm)	Sa (3 mm)
1	0.50	50	20	8.96	9.11	9.89
2	1.00	50	20	8.57	6.34	9.76
3	1.00	20	50	8.40	6.34	9.11
4	0.75	80	80	8.40	8.11	8.53
5	0.75	50	50	6.73	12.40	8.76
6	0.75	20	20	8.40	8.20	9.10
7	0.75	20	80	6.63	9.47	7.59
8	0.75	80	20	7.16	9.29	7.12
9	0.75	50	50	7.25	11.30	9.20
10	0.75	50	50	6.98	10.70	8.92
11	0.75	50	50	6.63	11.20	8.50
12	0.75	50	50	6.74	11.40	9.20
13	1.00	80	50	12.30	6.64	9.49
14	0.50	50	80	7.21	6.93	9.48
15	0.50	20	50	9.35	8.27	9.75
16	1.00	50	80	11.40	7.26	9.49
17	0.50	80	50	7.32	7.64	8.53

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having power density in a 16–32 MW/mm^2 range, remelting took place optimally producing low surface roughness.

The BBD experimental results of Sa (leveled) for all three laser beam diameters are tabulated in Table 3. From the table, it can be seen that for 1-mm spot diameter, the Sa value was found to be the minimum for laser energy 0.75 J and 50% overlap in both *X* and *Y* directions (corresponding to 6 J/mm^2 energy density). These process parameters are the mid-values, which are repeated 5 times. The average Sa value is 6.87 µm with a 0.22 standard deviation. For the 2-mm spot, the Sa value was found to be the minimum (6.34 µm) with a higher laser energy of 1 J and 20–50% overlap in the *X* and *Y* directions corresponding to an energy density of 1.88 J/mm^2. For the 3-mm spot diameter, the lowest Sa value was found to be 7.12 µm, higher than the other two spot diameters for 0.75 J laser energy and 80% *X* and 20% *Y* overlaps corresponding to 0.73 J/mm^2.

4 Process optimization by statistical analysis

4.1 Analysis of variance (ANOVA)

It is necessary to find out the most important parameters affecting the process (Table 4). It can be done using analysis of variance (ANOVA). The *P* and *F* test is used to validate the relevance of process parameters in ANOVA analysis. Evidence against the null hypothesis strengthens as the *P*-value decreases. A parameter or model's statistical significance may be determined using the *F*-value. Therefore, the most important parameter is the one with the highest *F*-value and the lowest *P*-value [33].

The statistical significance of each process parameter and its impact on surface roughness was examined using an ANOVA analysis. Prior to analysis, graphical analysis was used to successfully test the hypothesis of the normality of the residual distribution. A 95% degree of confidence ($\alpha = 0.05$) was used. Therefore, if the *P*-value was less than 0.05, a controlling factor or a set of control factors was deemed significant. Additionally, the *F*-value, which denotes the contribution to the factor's response variability, was considered. It is evident from the results for the quadratic model of a 1 mm diameter beam that the main process parameters affecting the surface roughness are laser energy, interaction between laser energy and overlap *X*, interaction between laser energy and overlap *Y*, interaction between overlap *X* and *Y*, and all the square terms except overlap *Y* square. For the 2-mm diameter beam, the significant factors are laser energy, interaction between laser energy and overlap *Y*, interaction between overlap *X* and *Y*, and all square terms. In the case of 3-mm diameter beam laser polishing, it is found that overlap *X*, interaction between overlap *X* and *Y*, and all the square terms except overlap *Y* square are significant. Squared terms (A^2, B^2, C^2) for laser energy and overlaps (*X*%, *Y*%) capture nonlinear effects and threshold behavior in laser polishing. They reveal complex relationships between process parameters and surface roughness, aiding optimization for enhanced component quality.

4.1.1 Pareto analysis and normal probability plots

The Pareto plot makes it possible to identify the factors and interactions that have the greatest impact on a process or design optimization research. It draws a reference line on the graph and shows the absolute values of effects. Any influence that crosses this reference line has the potential to be significant. It can also be used for a comparison between different input process parameters and their interactions for their degree of significance in affecting the output response. The significance level chosen here is 0.05. From the Pareto graph, it is evident that the most important interaction for Sa (1 mm), Sa (2 mm), and Sa (3 mm) is the square of laser energy. The comparison between other factors for different spot diameters can be obtained by considering the bar graph. The normal probability plot for all three cases considered is provided in Fig. 13, where it can be observed that the residuals are distributed along the predicted normal line. Thus, it can be considered that the assumption that the residuals are normally distributed is true.

Table 4 ANOVA results in terms of *F*-value and *P*-value

Source	Sa (1 mm)		Sa (2 mm)		Sa (3 mm)	
	***F*-value**	***P*-value**	***F*-value**	***P*-value**	***F*-value**	***P*-value**
A (laser energy)	49.30	0.0002	14.03	0.0072	0.7686	0.4097
B (*X*% overlap)	4.63	0.0684	0.1752	0.6881	13.88	0.0074
C (*Y*% overlap)	0.2432	0.6370	0.6661	0.4413	1.14	0.3208
AB	56.55	0.0001	0.8417	0.3894	2.17	0.1844
AC	33.73	0.0007	9.35	0.0184	0.0736	0.7941
BC	14.57	0.0066	5.84	0.0463	32	0.0008
A^2	106.29	0.0001	125.53	0.0001	43.86	0.0003
B^2	9.54	0.0176	32.58	0.0007	34.33	0.0006
C^2	2.22	0.1802	24.49	0.0017	0.5350	0.4883

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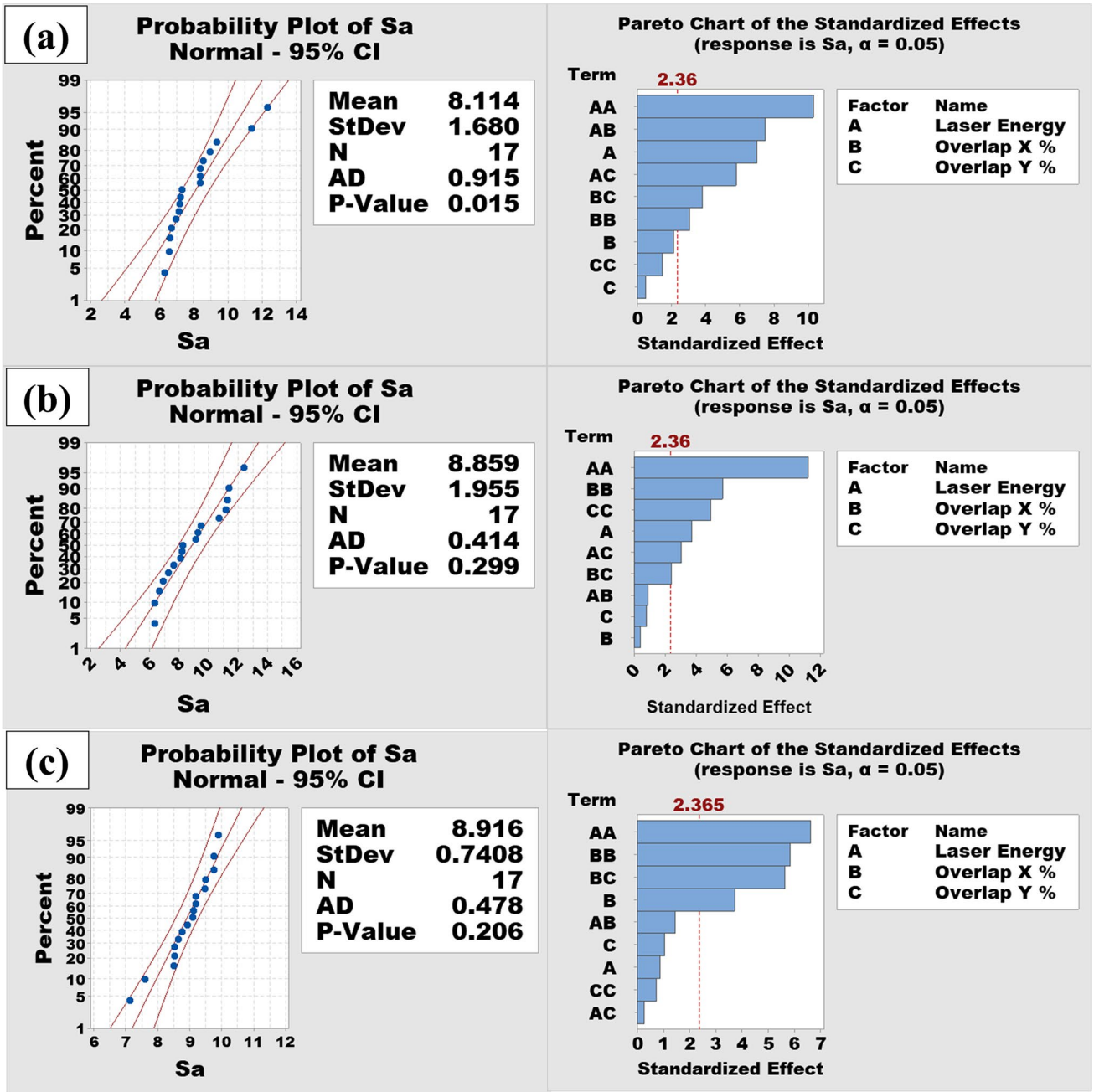


Fig. 13 Normal probability plot for residuals and Pareto chart of standardized effects. **a** Sa (1 mm). **b** Sa (2 mm). **c** Sa (3 mm)

4.2 Effect of process parameters on the response

Surface polishing by multiple overlapped laser pulses of 10 ns pulse duration and power density in the MW-GW/mm^2 range is a complex process involving a number of phenomena, including remelting, melt flow, evaporation, ablation, melt ejection, and spatter recast, which are influenced by changes in viscosity and surface tension with temperature, recoil pressure of vapor, buoyancy, and gravity [34]. These are influenced not only by the individual process parameters but also by their interactions with each other. Therefore, explaining the results of surface roughness for varying parameters is not always straightforward. However, some of the plausible phenomena involved in a single laser pulse interaction are briefly presented in Table 5 for different laser power densities of varying spot diameters. Based on these phenomena, an attempt will be made to explain the effects of variation of different process parameters on the surface roughness analyzed by contour and 3D surface plots presented in the following sections. It can also suggest

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Table 5 Plausible phenomena involved in pulsed laser polishing for different laser powder densities

Laser Spot Diameter (mm)	Laser Pulse Energy (J)	Laser Power Density (MW/mm²)	Temperature & Viscosity	Laser Irradiation Effects	Morphology of laser irradiated zone Schematic	Changes in Surface Roughness
1	1	127	Increasing Temperature / Decreasing viscosity	Over melting+Ablation Melt ejection+ Spatter formation at periphery Drilling effect		---
1	0.75	96		Over melting+Evaporation Melt ejection+ Spatter formation at periphery Drilling effect		+/-
1	0.5	64		Over melting+Evaporation Melt ejection+ Recast Drilling effect		-
2	1.0	32		Melting + Evaporation Recoil vapour pressure causing radially outward melt flow + Marangoni flow radially inward due to positive surface tension temperature coefficient		++
2	0.75	24		Melting + Onset of evaporation Recoil vapour pressure causing radially outward melt flow forming recast layer at the periphery		--
2	0.5	16		Melting+ Melt flow from peaks to valleys		+
3	1.0	14		Melting+ radially outward melt flow		+/-
3	0.75	11		Melting of peaks+ Onset of melt flow		+
3	0.5	7		Shallow & partial melting		--

what may be the region of interest where one can get a good surface finish by the combination of different process parameters.

4.2.1 Case 1: 1-mm diameter laser beam

From the contour and surface plots (Fig. 14a–d), it is evident that the Sa value is high at the counterpoints for the extreme combinations of laser power and overlap, the highest being for the high values of laser energy and overlaps.

The lowest value of Sa is observed for the medium levels of the laser energy and overlap %. A similar trend is observed in Fig. 14c and f with the variation of % overlaps. As described earlier, in the case of a 1-mm spot diameter, the over-melting and drilling effects are the more probable causes for the higher Sa values, specially at 1 J laser energy because of very high-power density (Table 5). For the medium levels of laser energy and % overlap, the surface roughness could get even out resulting in a lower Sa value.

4.2.2 Case 2: 2-mm diameter laser beam

For the 2-mm diameter beam case, the trend for all the plots as shown in Fig. 15 is the same. All corner points (extreme laser energy and % overlaps) provide the lowest Sa values, whereas the medium-level combinations provide the highest value of Sa. In Fig. 15a and d, the lowest Sa is obtained for the high laser energy and the low overlap *X*%. Similarly, in Fig. 15b and e, the lowest Sa is obtained for high laser energy and low overlap *Y*%. Further, the lowest Sa value is obtained for low overlap *X*% and low overlap *Y*% as seen in Fig. 15c and f. As indicated in Table 5, for the lower laser pulse energy, melting and melt flow from peaks to valleys take place, thereby reducing the surface roughness. For the medium laser energy, the recoil pressure of the onset of evaporation could create a valley pushing the molten material radially outward, thereby causing an increase in the surface roughness. The surface tension temperature coefficient of CoCrMo alloy reportedly has a boomerang shape, i.e., surface tension, first increases with temperature unto 1800 K and then tends to decrease with further increase of temperature [35]. At the higher laser energy, the radially inward Marangoni flow could counteract the outward melt flow due to recoil pressure, thus creating a more even polished surface [36].

4.2.3 Case 3: 3-mm diameter laser beam

For the 3-mm beam diameter case, the trend for the plots in Fig. 16a, b, d, and e is the same. Low Sa values are obtained for the middle value of laser energy due to the melting of peaks and the onset of melt flow (Table 5). The highest Sa value lies in the region of low and high laser energy but medium overlap *X*% or overlap *Y*%. While the partial and shallow melting at low laser energy could be the reason for the highest Sa value, at higher laser energy, this could be due to the radially outward melt flow. In Fig. 16c and f, the

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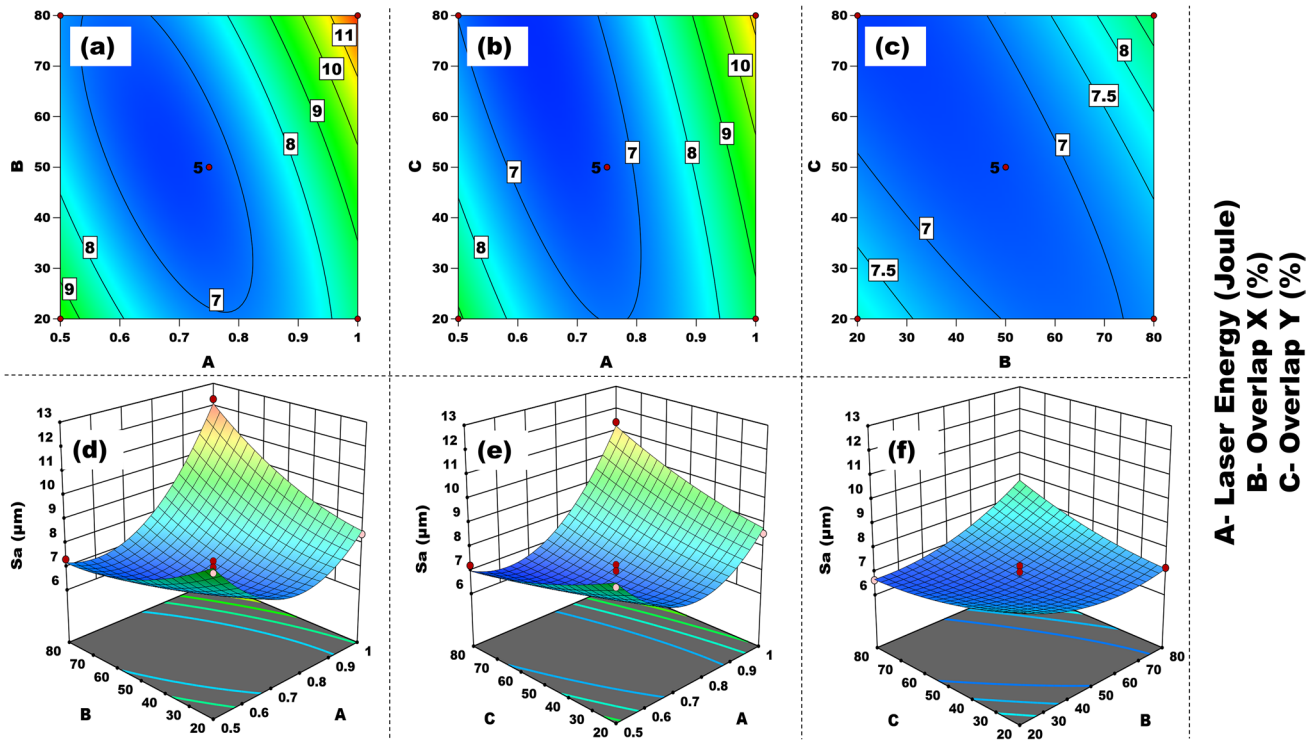


Fig. 14 Effect of different process parameters on the surface characteristics. **a** Contour plot between overlap *X*% and laser energy. **b** Contour plot between overlap *Y*% and laser energy. **c** Contour plot between overlap *X*% and overlap *Y*%. **d** 3D surface plot between overlap *X*% and laser energy. **e** 3D surface plot between overlap *Y*% and laser energy. **f** 3D surface plot between overlap *X*% and overlap *Y*%

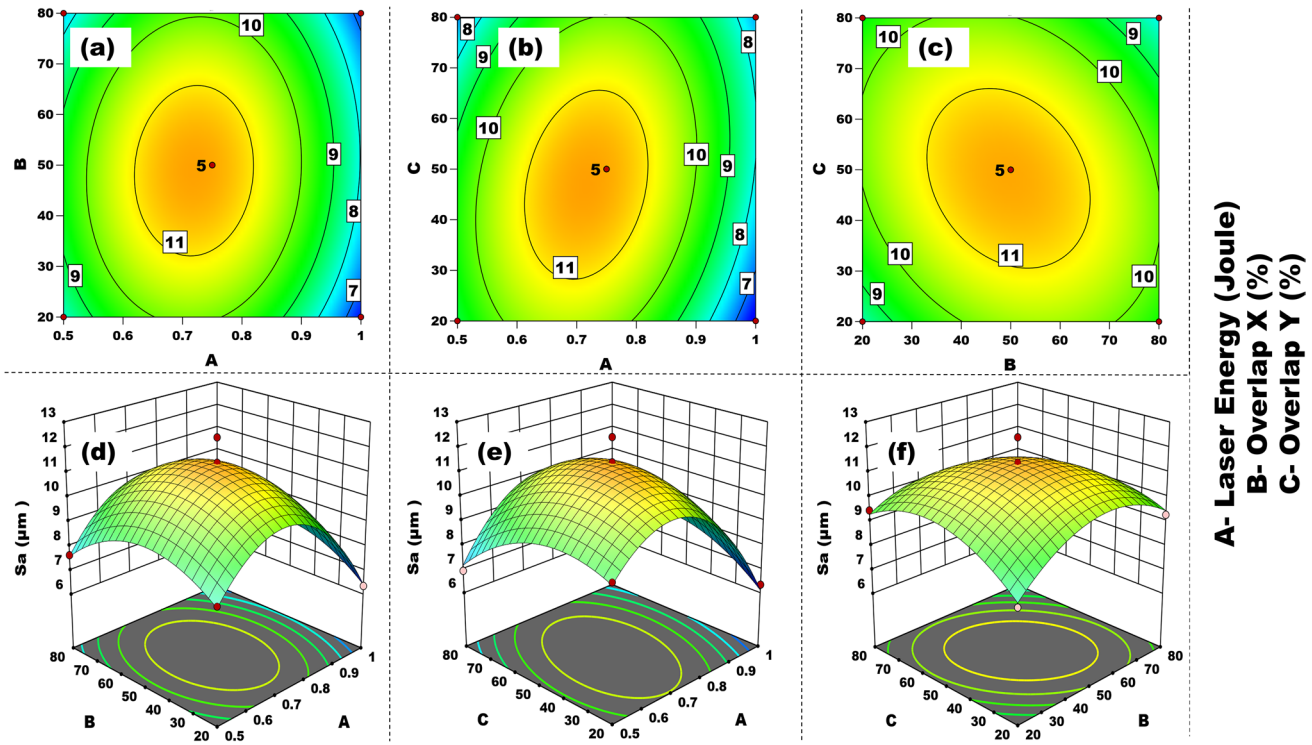


Fig. 15 Effect of different process parameters on the surface characteristics. **a** Contour plot between overlap *X*% and laser energy. **b** Contour plot between overlap *Y*% and laser energy. **c** Contour plot between overlap *X*% and overlap *Y*%. **d** 3D surface plot between overlap *X*% and laser energy. **e** 3D surface plot between overlap *Y*% and laser energy. **f** 3D surface plot between overlap *X*% and overlap *Y*%

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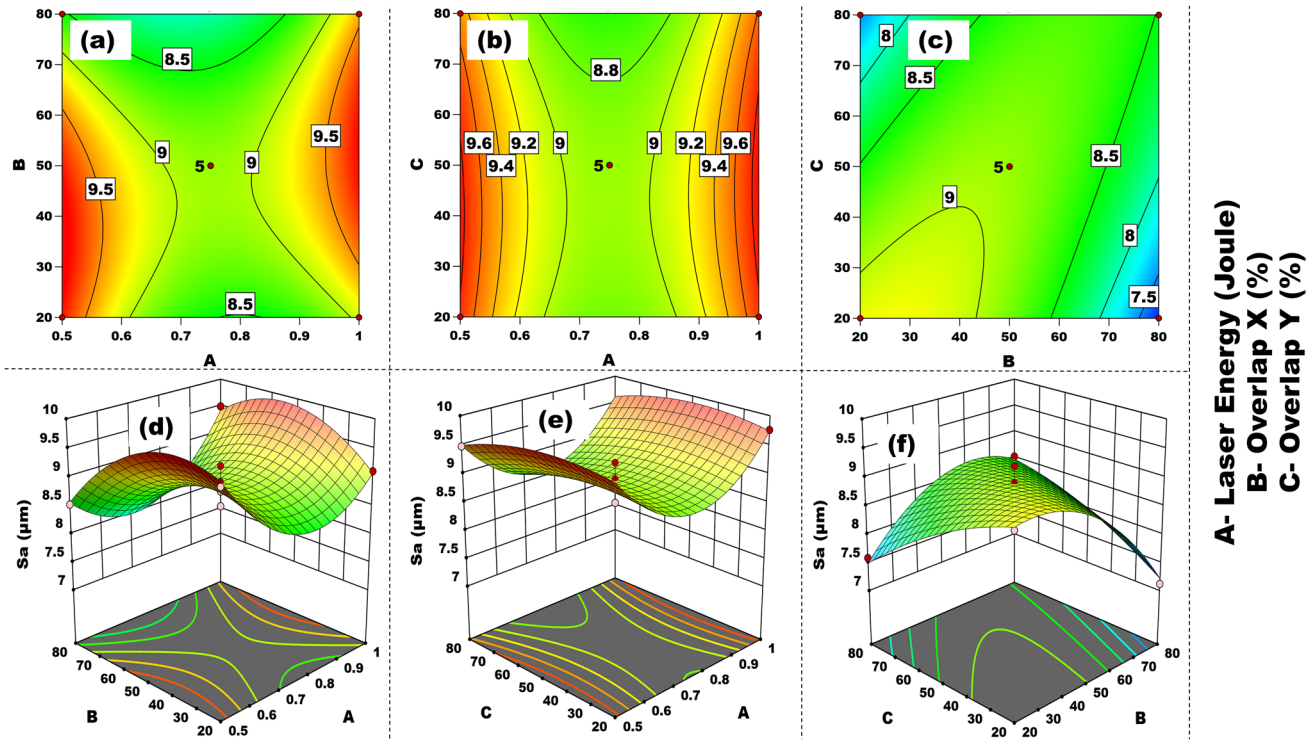


Fig. 16 Effect of different process parameters on the surface characteristics. **a** Contour plot between overlap *X*% and laser energy. **b** Contour plot between overlap *Y*% and laser energy. **c** Contour plot between overlap *X*% and overlap *Y*%. **d** 3D surface plot between overlap *X*% and laser energy. **e** 3D surface plot between overlap *Y*% and laser energy. **f** 3D surface plot between overlap *X*% and overlap *Y*%

lowest value of Sa lies in two regions of high and low combination of *X*% and *Y*% overlaps. The highest value of Sa lies in the region of 20–40% overlap *X* and 20–30% overlap *Y* due to partial-melting of the partial melting of peak and valley due to less energy density.

In a regression model, *R*-squared (also known as R^2 or the coefficient of determination) is a statistical metric that quantifies how much of the variance in the dependent variable can be accounted for by the independent variable. *R*-squared, thus, displays how well the data match the regression model (the goodness of fit). From the fit statistics for all the responses in Table 6 (Sa of 1 mm, 2 mm, and 3 mm), it is found that there is a good agreement between the predicted R^2 and the adjusted R^2 having a difference less than that of a value of 0.2 suggesting the utilization of the regression model so obtained for predicting the output or response with good accuracy.

Table 6 Fit statistics tabulation

Responses	Std. Dev	Mean	R^2	Adjusted R^2	Predicted R^2
Sa (1 mm)	0.3943	8.11	0.9759	0.9449	0.7849
Sa (2 mm)	0.5068	8.86	0.9706	0.9328	0.8931
Sa (3 mm)	0.2581	8.92	0.9469	0.8786	0.7308

4.3 Process parameter optimization by RSM

To identify the ideal process conditions, the Response Surface Method (RSM) and Response Optimization (R.O.) were used. RSM provides a regression model that illustrates how the process parameters affect the surface properties. The model uses ten constants to relate the response variables (Sa (1 mm), Sa (2 mm), and Sa (3 mm)) to the process variables (*A*, *B*, and *C*), their interactions ($A \times B$, $A \times C$, and $B \times C$), and square terms (A^2, B^2 and C^2). These regression models can be utilized for prediction and obtaining the optimum values of process parameters as per the required objective function. These models can also be utilized for predicting the global optimum values using different metaheuristic algorithms (G.A., PSO, TLBO, etc.). By comparing the factor coefficients, these regression models can be used to determine the relative importance of the factors involved which is given in supplementary data S11.

The optimization is performed utilizing Minitab statistical software by the approach of estimating response desirability (*d*) in RSM. The following Eqs. 1, 2, and 3 are used to determine the desirability of minimizing a single response:

$$d = 0, if R > UB \quad (1)$$

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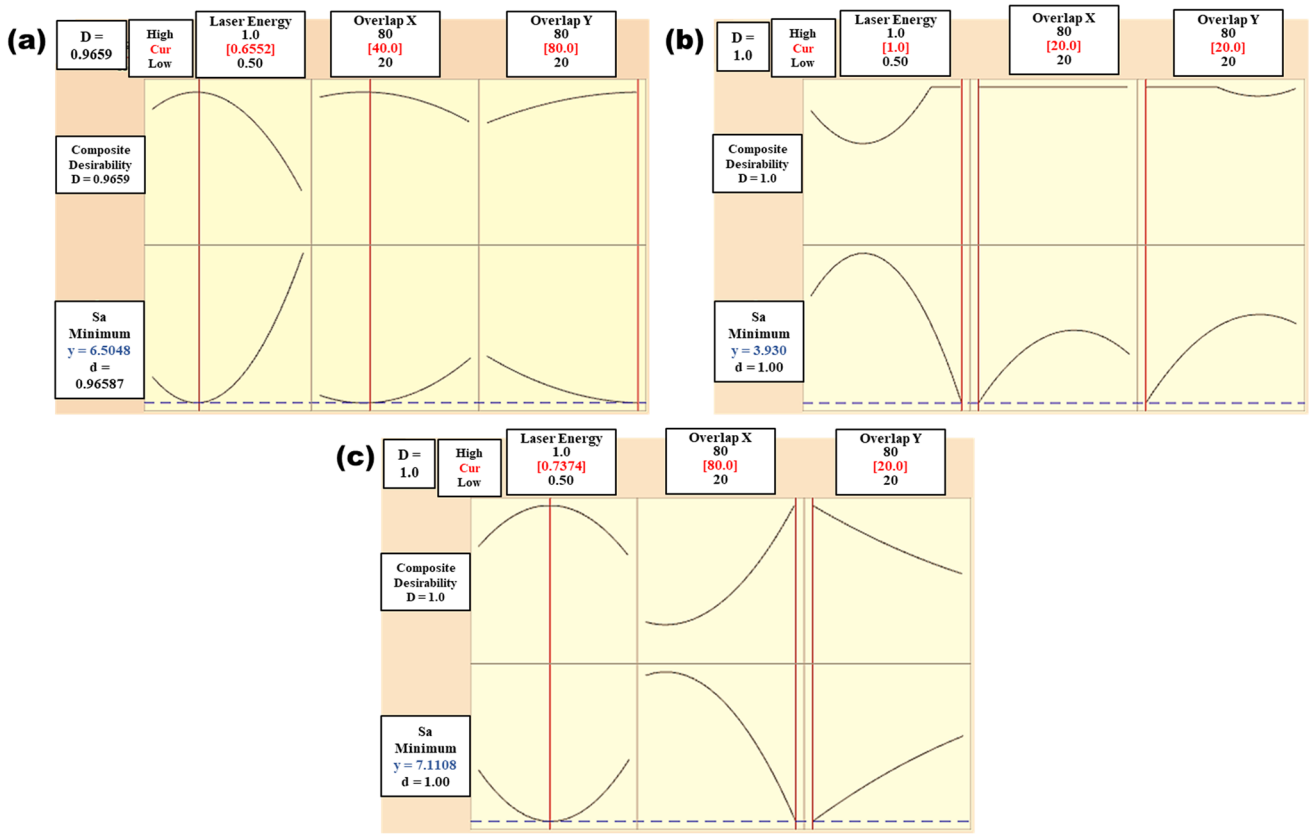


Fig. 17 Different Optimum solutions provided by response optimizer using RSM with high desirability. **a** Sa (1 mm). **b** Sa (2 mm). **c** Sa (3 mm)

$$d = \left(\frac{UB - R}{UB - T}\right)^{w}, if T < R < UB \tag{2}$$

$$d = 1, if R < T \tag{3}$$

Here, d is the desirability, R is the response value, T is the target value, UB is the value of the upper bound, and w is the weight defined as the desirability function. The desirability function is utilized for considering the response for its desirable value calculation in the range of 0 to 100%. The target value represents the desired value; like for Sa, it should be as less as possible. Weight is used for the distribution of the desirability function between the target and the upper bound. The upper bound is the highest permissible value that can be achieved by the response; it depends upon the experimental data provided and can also be chosen as per one's choice. When there is little difference between the target and the upper bound values, desirability increases. Figure 17 shows different optimum solutions provided by the response optimizer using RSM with high desirability for various laser spot size diameters.

Here for various input process parameters and for the response surface roughness (Sa) for each of the cases along with the regression models, a desirability value is calculated by the desirability function. The values of the T, UB, and w are provided in Table 7.

Table 7 Optimized results for various laser spot diameters considered

Responses	Laser energy (A)	Overlap X% (B)	Overlap Y% (C)	Sa (predicted)	Desirability (%)	Target (T)	Upper bound (UB)	Weight (w)
Sa (1 mm)	0.655	37.578	80	6.50	96.65%	6.63	12.3	1
Sa (2 mm)	1	20	20	3.93	100%	6.34	12.4	1
Sa (3 mm)	0.737	80	20	7.11	100%	7.12	9.89	1

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5 Conclusions

From the experimental studies and statistical analysis performed on surface polishing of L-PBF-built CoCr alloy samples by multiple overlapped laser pulses of varying laser energy, laser spot diameter, and overlap rates, the following observations are made:

1 Pulsed laser remelting was able to reduce the surface roughness of as-built LPBF CoCr surfaces significantly. Maximum AQreduction in surface roughness was obtained experimentally with the laser spot of 2 mm diameter, 1 J laser energy, and overlaps of 20% X and 50% Y as well as 50% X and 20% Y. While the reduction in surface roughness was limited by the drilling-like effect in the case of 1-mm diameter laser spot due to high laser power densities, it was due to the incomplete melting in the case of 3mm diameter laser spot.
2 From the ANOVA performed with the P and F test of the laser polishing parameters, the significant parameter which affects the polishing of CoCr material was found to be the square of the laser power.
3 The Pareto chart was able to predict the significant parameter as the square of the laser power for all the laser spot diameters. Normal probability plots with residuals along the predicted lines suggest a correct interpretation of residuals.
4 The regression model so obtained for predicting the output or response (Sa) shows good accuracy of the model.
5 The optimized results obtained using response optimizer (RSM) are laser energy of 0.65 J, overlap X% of 37.57%, and overlap Y% of 80% for 1-mm spot diameter providing a Sa value of 6.5. For a 2-mm spot diameter, the optimized value of laser energy is 1 J, overlap X% of 20%, and overlap Y% of 20% providing a Sa value of 3.93. For a 3-mm spot diameter, the optimized value of laser energy is 0.737 J, overlap X% of 80%, and overlap Y% of 20% providing a Sa value of 7.11. Considering the optimum values obtained for various spot diameters for the reduction of surface roughness, it has been found that a 2-mm spot diameter shows the best result.
6 With energy density, the micro-hardness of the polished surfaces varied around the base-material hardness for all the energy densities, which could be attributed to the modification in microstructure morphology and microstructure feature size.

Due to the relatively low depth of modification produced by nanosecond order laser pulses, the pulse laser micro-polishing could only be employed to reduce the peaks and valleys on the surface of as-built L-PBF parts. Also, due to the small spot sizes for the current study, limited by the maximum pulse energy of the laser, the process may be time-consuming. A future study involving higher laser energy and larger spot size has the potential to overcome these limitations. In the present research, a DoE- and RSM-based approach has been utilized for parametric optimization of the surface roughness. In future research, more analysis on the metallurgical and tribological properties of the laser-polished surface can be considered.

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Author contribution Abhishek Kumar: conceptualization, investigation, formal analysis, methodology, validation, writing—original draft. Harikrishnan Ramadas: investigation, methodology, writing—review and editing. Bijay Bikram Samal: formal analysis, methodology, visualization, data curation, writing—review and editing. Cheruvu Siva Kumar: supervision, resources, funding acquisition, writing—review and editing. Ashish Kumar Nath: conceptualization, project administration, supervision, resources, funding acquisition, writing—review and editing.

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Data availability The raw/processed data required to reproduce the experimental findings cannot be shared at this time as the data also forms part of an ongoing study.

Declarations

Ethics approval The authors declare that this manuscript has not been published elsewhere and is not under consideration by another journal.

Consent to participate All authors have approved the manuscript and agree with its submission.

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Competing interests The authors declare no competing interests.

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